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The de Rham cohomology of covers with a cyclic p -Sylow subgroupJędrzej Garnek^a, Aristides Kontogeorgis^b^a Faculty of Mathematics and Computer Science, Adam Mickiewicz University, ul. Uniwersytetu Poznańskiego 4, Poznań, 61-614, Poland^b Department of Mathematics, National and Kapodistrian University of Athens, Panepistimioupolis, 15784, Athens, Greece

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ABSTRACT

Let X be a smooth projective curve over a field k with an action of a finite group G . A well-known result of Chevalley and Weil describes the $k[G]$ -module structure of cohomologies of X in the case when the characteristic of k does not divide $\#G$. It is unlikely that such a formula can be derived in the general case, since the representation theory of groups with non-cyclic p -Sylow subgroups is wild in characteristic p . The goal of this article is to show that when G has a cyclic p -Sylow subgroup, the G -structure of the de Rham cohomology of X is completely determined by the ramification data. In principle, this leads to new formulas in the spirit of Chevalley and Weil for such curves. We provide such an explicit description of the de Rham cohomology in the cases when $G = \mathbb{Z}/p^n$ and when the p -Sylow subgroup of G is normal of order p .

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1. Introduction

In the 1930s Chevalley and Weil gave an explicit description of the equivariant structure of the module of holomorphic differentials of a smooth projective curve X with an action of a finite group G over a field of characteristic 0 (cf. [8]). Their formula depends on the genus of the quotient curve X/G and on the ramification data of the quotient morphism $X \rightarrow X/G$. This result extends to the case of the sheaf cohomologies of line bundles on a curve over a field of characteristic $\text{char } k \nmid \#G$ (cf. [11]) and can be easily used to derive a formula for the de Rham cohomology of a curve. However, it is hard to expect such a formula for all finite groups. Indeed, if G is a group with a non-cyclic p -Sylow subgroup, the set of indecomposable $k[G]$ -modules is infinite. If, moreover, $p > 2$ then the indecomposable $k[G]$ -modules are considered impossible to classify (cf. [25]). There are many results concerning the equivariant structure of cohomologies for particular groups (see e.g. [29] for the case of cyclic groups, [24] for abelian groups, [5] for groups with a cyclic Sylow subgroup, or [4] for the Klein group) or curves (cf. [23], [10], [16], [22]). Also, one may expect that (at least in the case of p -groups) determining cohomologies comes down to Harbater–Katz–Gabber covers (cf. [13], [14]). However, there is no hope of obtaining a result similar to the one of Chevalley and Weil.

This brings attention to groups with a cyclic p -Sylow subgroup. For those, the set of equivalence classes of indecomposable modules is finite (cf. [19], [7], [18]). While their representation theory still seems a bit too complicated to derive a general formula for the cohomologies, the article [5] shows that the $k[G]$ -module structure of $H^0(X, \Omega_X)$ is determined by the genus of X and *ramification data*, that is:

- the stabilizers of the action on $X(\bar{k})$,
- the lower (or equivalently, upper) ramification jumps of the action corresponding to the stabilizers,
- the fundamental characters of the action (see Section 2 for the definition).

The main result of this article is a similar statement for the de Rham cohomology.

Main Theorem. *Suppose that G is a group with a p -cyclic Sylow subgroup. Let X be a curve with an action of G over a perfect field k of characteristic p . The $k[G]$ -module structure of $H_{dR}^1(X)$ is uniquely determined by the ramification data of the cover $X \rightarrow X/G$ and the genus of X .*

In order to prove Main Theorem, we show explicit formulas for the G -structure of $H_{dR}^1(X)$ when $G \cong \mathbb{Z}/p^n$ (cf. Theorem 4.1) and when G has a normal Sylow subgroup of order p (cf. Theorem 6.1). In principle, the proof of Main Theorem could be made effective to give an explicit formula for $H_{dR}^1(X)$ for an arbitrary group with a cyclic p -Sylow subgroup. This seems however really complicated for several reasons. Firstly, even though the indecomposable modules for groups with a cyclic p -Sylow subgroup can

be explicitly described (cf. [20]), this description is pretty involved. Secondly, already in the case when $G \cong \mathbb{Z}/p \rtimes_{\chi} C$, $p \nmid \#C$, the explicit formula is quite long.

We elaborate now on the proof of Main Theorem. The first step is to provide an explicit formula in the case when $G = \mathbb{Z}/p^n$. This result is proven by applying induction on n twice: once for the curve X with an action of $\mathbb{Z}/p^{n-1}$ and once for the curve $X'' := X/(\mathbb{Z}/p)$. The second step is to describe $H_{dR}^1(X)$ in case when $G \cong \mathbb{Z}/p \rtimes_{\chi} C$ and $p \nmid \#C$. The proof in this case follows by analyzing the $\mathbb{Z}/p$ -invariants in the Hodge–de Rham exact sequence. Then we prove the result for groups of the form $\mathbb{Z}/p^n \rtimes_{\chi} \mathbb{Z}/c$ by using induction similarly as in the first step. Finally, we use Conlon induction theorem to reduce Main Theorem to the case when G is of the form $\mathbb{Z}/p^n \rtimes_{\chi} \mathbb{Z}/c$.

Note that if $p > 2$ and the p -Sylow subgroup of G is not cyclic, the structure of $H_{dR}^1(X)$ isn't determined uniquely by the ramification data. Indeed, see [15] for a construction of G -covers with the same ramification data, but varying $k[G]$ -module structure of $H^0(X, \Omega_X)$ and $H_{dR}^1(X)$.

As noted by the reviewer, the methods used in the proof of Main Theorem allow to describe the structure of the de Rham cohomology only over perfect fields. Over non-perfect fields, the definition of the ramification groups is quite different and goes back to Abbes and Saito (cf. [1]). It is an interesting question whether, in this context, the Main Theorem still holds when the “ramification data” refers to these ramification groups.

Outline of the paper In Section 2 we discuss notation and preliminaries. Section 3 contains several results concerning modular representation theory. In Section 4 we show the formula for the de Rham cohomology of $\mathbb{Z}/p^n$ -covers. The next Section is devoted to the proof of Main Theorem, assuming the result for groups with a p -Sylow subgroup of order p . The formula for such groups is proven in Section 6. In the final section we provide an explicit example of curves with an action of $G := \mathbb{Z}/p \rtimes_{\chi} \mathbb{Z}/(m \cdot (p-1))$ and compute the G -structure of their de Rham cohomologies.

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2. Notation and preliminaries

Assume that $\pi : X \rightarrow Y$ is a G -cover of smooth projective curves over a perfect field k of characteristic p . Throughout the paper we will use the following notation for any $P \in X(\bar{k})$:

- G_P is the stabilizer group at P ,
- $e_{X/Y,P}$ is the ramification index at P ,
- $m_{X/Y,P} := \text{ord}_p(e_{X/Y,P})$ is the maximal power of p dividing the ramification index,
- $m_{X/Y} := \max\{m_{X/Y,P} : P \in X(k)\}$,
- $u_{X/Y,P}^{(t)}$ (resp. $l_{X/Y,P}^{(t)}$) is the t -th upper (resp. lower) ramification jump at P for $t \geq 1$,
- $u_{X/Y,P}^{(0)} := 1$ for any ramified point $P \in X(\bar{k})$ (note that this is not a standard convention),
- $u_{X/Y,P} := u_{X/Y,P}^{(m_{X/Y,P})}$ is the last ramification jump.

By Hasse–Arf theorem (cf. [27, p. 76]), if the p -Sylow subgroup of G is abelian, the numbers $u_{X/Y,P}^{(t)}$ are integers. For any $Q \in Y(\bar{k})$ we denote also by abuse of notation $e_{X/Y,Q} := e_{X/Y,P}$, $u_{X/Y,Q}^{(t)} := u_{X/Y,P}^{(t)}$, $G_Q := G_P$, etc. for arbitrary $P \in \pi^{-1}(Q)$. Note that G_Q is well-defined only up to conjugacy. Let

$$B_{X/Y} := \{Q \in Y(\bar{k}) : e_{X/Y,Q} > 1\}$$

be the ramification locus of π . We recall now the classical Chevalley–Weil formula. Assume that $k = \bar{k}$. Let $\theta_{X/Y,P} : G_P \rightarrow \text{Aut}_k(\mathfrak{m}_P/\mathfrak{m}_P^2) \cong k^\times$ be the *fundamental character of P* . In other words, if t_P is a uniformizer at P then for any $\sigma \in G_P$:

$$\theta_{X/Y,P}(\sigma) \equiv \frac{\sigma(t_P)}{t_P} \pmod{\mathfrak{m}_P^2}.$$

Again, for $Q \in Y(k)$ we write $\theta_{X/Y,Q} := \theta_{X/Y,P}$ for any $P \in \pi^{-1}(Q)$. For any indecomposable $k[G]$ -module W we denote by $N_{Q,i}(W)$ the multiplicity of the character $\theta_{X/Y,Q}^i$ in the $k[G_Q]$ -module $W|_{G_Q}$. In the sequel we often use the Iverson bracket notation:

$$\llbracket \mathscr{P} \rrbracket = \begin{cases} 1, & \text{if the statement } \mathscr{P} \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

Proposition 2.1. *Keep the above notation. Assume additionally that $p \nmid \#G$ and $k = \bar{k}$. Let also $D = \sum_{P \in X(k)} m_P \cdot P \in \text{Div}(X)$ be an effective G -equivariant divisor. Write $m_Q := m_P$ for any $P \in \pi^{-1}(Q)$. Then:*

$$H^0(X, \Omega_X(D)) \cong \bigoplus_{W \in \text{Indec}(k[G])} W^{\oplus a(W)}, \quad (2.1)$$

where:

$$\begin{aligned} a(W) := & \left((g_Y - 1) + \frac{\deg D}{\#G} \right) \cdot \dim_k W \\ & + \sum_{Q \in Y(k)} \sum_{i=1}^{e_{X/Y,Q}-1} \left\langle \frac{-m_Q - i}{e_{X/Y,Q}} \right\rangle \cdot N_{Q,i}(W) + \llbracket D = 0 \rrbracket \cdot \llbracket W \cong k \rrbracket \end{aligned}$$

and $\langle x \rangle$ denotes the fractional part of x (i.e. $\langle x \rangle := x - \lfloor x \rfloor$).

Proof. For $D \neq 0$ this follows from [11, Theorem 3.8] by noting that $H^1(X, \Omega_X(D)) = 0$ and that the multiplicity of $\theta_{X/Y, Q}$ in

$$\Omega_X(D)_P / \mathfrak{m}_P \Omega_X(D)_P = \text{Span}_k(t_P^{-m_Q} dt_P)$$

equals $1 - m_Q$ for $P \in \pi^{-1}(Q)$. For $D = 0$ the same reasoning applies by noting that $H^1(X, \Omega_X) \cong k$ is a trivial $k[G]$ -module. $\square$

Proposition 2.1 and Serre's duality imply that under the above assumptions the $k[G]$ -module structure of $H^1(X, \mathcal{O}_X(-D)) \cong H^0(X, \Omega_X(D))^\vee$ is uniquely determined by the ramification data. Note also that the de Rham cohomology of X satisfies the Hodge–de Rham exact sequence:

$$0 \rightarrow H^0(X, \Omega_X) \rightarrow H_{dR}^1(X) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow 0. \quad (2.2)$$

Thus, since the category of $k[G]$ -modules is semisimple under the above assumptions, the ramification data determines the structure of $H_{dR}^1(X)$ as well.

3. Modular representation theory

In this section we review the modular representation theory in characteristic p for a group G with a normal cyclic p -Sylow subgroup H . By the Schur–Zassenhaus theorem (cf. [26, Theorem 7.41]), G is of the form:

$$G \cong H \rtimes_\chi C \quad \text{with } H \cong \mathbb{Z}/p^n, p \nmid \#C \text{ and } \chi : C \rightarrow \text{Aut}(H). \quad (3.1)$$

Let $H = \langle \sigma \rangle$. Observe that since $|\text{Aut}(H)| = p^{n-1} \cdot (p-1)$, χ must factor through $\text{Aut}(\mathbb{Z}/p) \cong (\mathbb{Z}/p)^\times$. Thus we can identify χ with a homomorphism $\chi : C \rightarrow \mathbb{F}_p^\times \subset k^\times$. Under this identification, we have $\rho\sigma\rho^{-1} = \sigma^{\chi(\rho)}$ for any $\rho \in C$. The representation theory of groups of the form (3.1) is well-understood. Assume that k is algebraically closed of characteristic p . Let $\text{Indec}(k[G])$ be the set of isomorphism classes of indecomposable $k[G]$ -modules. For every $k[G]$ -module U , we will denote by U_H the $k[H]$ -module with restricted action. By [3, Lemma 8, p. 34] we have that $\text{rad}(U) = \text{rad}(U_H) = (1 - \sigma)U$. Hence the socle of U (i.e. $\text{soc}(U) := \{u \in U : \text{rad}(k[G])u = 0\}$) consists of the elements $u \in U$ such that $(1 - \sigma)u = 0$; that is $\text{soc}(U) = U^\sigma$.

Every indecomposable module U is uniserial by [3, p. 42] and is characterized by the composition length and the simple module $U/\text{rad}(U) = S$. This means that there is a unique indecomposable projective module P that corresponds to S , i.e. $P/\text{rad}(P) = S$, [3, Theorem 3, p. 31]. Then by [3, Lemma 5, p. 33] we have that U is a homomorphic image of P . Moreover $\text{soc}(P) = \text{soc}(U) = S$ by [3, Proposition 5, p. 6]. This proves that $S = U/\text{rad}U = P/\text{rad}(P) = \text{soc}(P) = \text{soc}(U)$, by [3, Theorem 6, p. 43].

Assume that $U^\sigma = S$ has dimension $\dim_k U^\sigma = d$. Then by [3, p. 35] we have that each successive quotient of P and U is d -dimensional. This proves that the decomposition length equals $\dim_k U / \dim_k U^\sigma$.

We summarize the above discussion in the following well-known result.

Proposition 3.1. *For $U \in \text{Indec}(k[G])$, the socle $U^\sigma = \ker(\sigma - 1)$ belongs to $\text{Indec}(k[C])$. The map*

$$\begin{aligned} \text{Indec}(k[G]) &\rightarrow \text{Indec}(k[C]) \times \{1, \dots, p^n\} \\ U &\mapsto \left(U^\sigma, \frac{\dim_k U}{\dim_k U^\sigma} \right) \end{aligned}$$

is a bijection.

We write $J_i(M)$ for the $k[G]$ -module corresponding to a pair $(M, i) \in \text{Indec}(k[C]) \times \{1, \dots, p^n\}$. Moreover, if V is $k[C]$ -module with the decomposition $V \cong \bigoplus_{M \in \text{Indec}(k[C])} M^{\oplus a_M}$ then we put:

$$J_i(V) := \bigoplus_{M \in \text{Indec}(k[C])} J_i(M)^{\oplus a_M}.$$

For any $k[H]$ -module M denote:

$$\begin{aligned} M^{(i)} &:= \ker((\sigma - 1)^i : M \rightarrow M), \\ T_H^i M &:= M^{(i)} / M^{(i-1)} \quad \text{for } i = 1, \dots, p^n. \end{aligned}$$

Note that if M is a $k[G]$ -module, then $T_H^i M$ is a $k[C]$ -module for $i = 1, \dots, p^n$. The modules $T_H^1 M, \dots, T_H^{p^n} M$ were crucial in the proof of [5, Theorem 1.1]. Their importance lies in the fact that they determine M completely (see Lemma 3.2 (5) below). We give now several facts concerning relations between those modules. To this end we need to introduce some notation. For any $k[C]$ -module M and any homomorphism $\psi : C \rightarrow k^\times$ we write $M^\psi := M \otimes_{k[C]} \psi$. In what follows, we fix n and write $T^i M := T_H^i M$. Let H' (respectively H'') be the unique subgroup (respectively quotient) of H isomorphic to $\mathbb{Z}/p^{n-1}$.

Lemma 3.2. *Let M be a $k[G]$ -module of a finite dimension.*

(1) *For any $1 \leq i \leq p^n - 1$ there exists a $k[C]$ -equivariant monomorphism:*

$$m_{\sigma^{-1}} : T^{i+1} M \hookrightarrow (T^i M)^{\chi^{-1}}.$$

(2) For any $1 \leq i \leq p^n - 1$ there exists an isomorphism of $k[C]$ -modules:

$$T_{H'}^i M \cong T^{pi-p+1} M \oplus T^{pi-p+2} M \oplus \dots \oplus T^{pi} M.$$

(3) If $\dim_k T_{H'}^i M = \dim_k T_{H'}^{i+1} M$ for some $1 \leq i \leq p^{n-1} - 1$ then

$$T^{j+1} M \cong (T^j M)^{\chi^{-1}} \quad \text{for } j = pi - p + 1, \dots, pi + p - 1.$$

(4) For any $j = 1, 2, \dots, p^{n-1}$ the trace

$$\mathrm{tr}_{\langle \sigma^{p^{n-1}} \rangle} := \sum_{i=0}^{p-1} (\sigma^{p^{n-1}})^i$$

induces a $k[C]$ -equivariant monomorphism:

$$T^{p^n - p^{n-1} + j} M \hookrightarrow T_{H''}^j M^{(p^{n-1})}.$$

(5) The $k[G]$ -structure of M is uniquely determined by the $k[C]$ -structure of $T^1 M, \dots, T^{p^n} M$.

Proof. (1) We define $m_{\sigma-1} : T^{i+1} M \rightarrow T^i M$ as follows:

$$m_{\sigma-1}(\overline{x}) := (\sigma - 1) \cdot x,$$

where for $\overline{x} \in T^{i+1} M$ we picked any representative $x \in M^{(i+1)}$. Indeed, if $x \in M^{(i+1)}$ then clearly $(\sigma - 1) \cdot x \in M^{(i)}$. Moreover $(\sigma - 1) \cdot x \in M^{(i-1)}$ holds if and only if $x \in M^{(i)}$. This shows that $m_{\sigma-1}$ is well-defined and injective. Thus it suffices to check that it is χ^{-1} -linear. Note that we have the following identity in the ring $k[G]$ for any $\rho \in C$:

$$(\sigma - 1) \cdot \rho = \rho \cdot (\sigma^{\chi(\rho)^{-1}} - 1) = \rho \cdot (\sigma - 1) \cdot (1 + \sigma + \sigma^2 + \dots + \sigma^{\chi(\rho)^{-1}-1})$$

Note that σ acts trivially on $T^i M$, so that for any $\overline{x} \in T^i M$:

$$(1 + \sigma + \sigma^2 + \dots + \sigma^{\chi(\rho)^{-1}-1}) \cdot \overline{x} = \chi(\rho)^{-1} \cdot \overline{x}.$$

This easily shows that

$$m_{\sigma-1}(\rho \cdot \overline{x}) = \chi(\rho)^{-1} \cdot \rho \cdot m_{\sigma-1}(\overline{x}),$$

which ends the proof.

(2) Note that $T_{H'}^i M = M^{(pi)}/M^{(pi-p)}$. The proof follows, since the category of $k[C]$ -modules is semisimple and $T_{H'}^i M$ has a filtration $M^{(pi-p+j)}/M^{(pi-p)}$ ($j = 1, \dots, p$) with $T^{pi-p+j} M$ as subquotients.

(3) By (1) and (2) we obtain:

$$\begin{aligned}\dim_k T_{H'}^i M &= \dim_k T^{p^i} M + \cdots + \dim_k T^{p^i-p+1} M \\ &\geq \dim_k T^{p^i+p} M + \cdots + \dim_k T^{p^i+1} M = \dim_k T_{H'}^{i+1} M.\end{aligned}$$

Since the left-hand side and right hand side are equal, we obtain:

$$\dim_k T^{p^i+p} M = \dim_k T^{p^i+p-1} M = \cdots = \dim_k T^{p^i-p+1} M.$$

The proof follows by part (1).

(4) Recall that in $\mathbb{F}_p[x]$ we have the identity:

$$1 + x + \cdots + x^{p-1} = (x - 1)^{p-1}.$$

Therefore in the group ring $k[H]$ we have:

$$\mathrm{tr}_{\langle \sigma^{p^{n-1}} \rangle} = \sum_{j=0}^{p-1} (\sigma^{p^{n-1}})^j = (\sigma^{p^{n-1}} - 1)^{p-1} = (\sigma - 1)^{p^n - p^{n-1}}.$$

This implies that:

$$\ker(\mathrm{tr}_{\langle \sigma^{p^{n-1}} \rangle} : M \rightarrow M^{(p^{n-1})}) = M^{(p^n - p^{n-1})}$$

and that $\mathrm{tr}_{\langle \sigma^{p^{n-1}} \rangle}$ induces the desired monomorphism. Similarly as in part (1) one shows that it is $\chi^{-(p^n - p^{n-1})}$ -linear. Finally, note that $\chi^{p^n - p^{n-1}} = 1$.

(5) This is basically [5, proof of Theorem 1.1]. We sketch the proof for reader's convenience. Write

$$M \cong \bigoplus_{i=1}^{p^n} \bigoplus_{W \in \mathrm{Indec}(C)} J_i(W)^{\oplus n(W,i)}.$$

Note that as $k[C]$ -modules:

$$T^j J_i(W) \cong \begin{cases} W^{\chi^{-j+1}}, & \text{if } j \leq i, \\ 0, & \text{if } j > i. \end{cases}$$

Hence:

$$T^j M \cong \bigoplus_{i=j}^{p^n} \bigoplus_{W \in \mathrm{Indec}(C)} (W^{\chi^{-j+1}})^{\oplus n(W,i)}$$

and the $k[C]$ -module structure of $T^j M$ determines uniquely the numbers:

$$\sum_{i=j}^{p^n} n(W, i)$$

for every $W \in \text{Indec}(k[C])$. This easily implies that the numbers $n(W, 1), \dots, n(W, p^n)$ are uniquely determined by the $k[C]$ -structure of $T^1 M, \dots, T^{p^n} M$. The proof follows. $\square$

If $G = H \cong \mathbb{Z}/p^n$ we abbreviate $J_i := J_i(k)$. Observe that J_i is given by the Jordan block of size i and eigenvalue 1. Moreover, for $i > 0$:

$$\dim_k T^i J_l = \llbracket i \leq l \rrbracket. \quad (3.2)$$

Lemma 3.2 (5) implies that the numbers $\dim_k T^i M$ for $i = 1, \dots, p^n$ determine the structure of a $k[H]$ -module M completely. If H' is as above, we denote the indecomposable $k[H']$ -modules by $\mathcal{J}_1, \dots, \mathcal{J}_{p^n-1}$.

4. Cyclic covers

The goal of this section is to prove the following formula for the de Rham cohomology of $\mathbb{Z}/p^n$ -covers of curves.

Theorem 4.1. *Let k be an algebraically closed field of characteristic p . Suppose that $\pi : X \rightarrow Y$ is a $\mathbb{Z}/p^n$ -cover. Pick an arbitrary $Q_0 \in Y(k)$ with $m_{X/Y, Q_0} = m_{X/Y}$. Then, as a $k[\mathbb{Z}/p^n]$ -module $H_{dR}^1(X)$ is isomorphic to:*

$$J_{p^n}^{2(g_Y-1)} \oplus J_{p^n-p^{n-m}+1}^2 \oplus \bigoplus_{\substack{Q \in B \\ Q \neq Q_0}} J_{p^n-p^n/e_Q}^2 \oplus \bigoplus_{Q \in B} \bigoplus_{t=0}^{m_Q-1} J_{p^n-p^{n+t}/e_Q}^{u_Q^{(t+1)}-u_Q^{(t)}}, \quad (4.1)$$

where we abbreviate $B := B_{X/Y}$, $e_Q := e_{X/Y, Q}$, $u_Q^{(t)} := u_{X/Y, Q}^{(t)}$, $m := m_{X/Y}$ and $m_Q := m_{X/Y, Q}$.

Remark 4.2. Note that this formula is well-defined for $g_Y = 0$, even though the first exponent is negative. Indeed, since $m_{X/Y} = n$ (as $\mathbb{P}^1$ doesn't have any étale covers), the first two summands in (4.1) cancel out.

We prove now several auxiliary facts used in the proof of Theorem 4.1.

Lemma 4.3. *If G is a finite group and the G -cover $\pi : X \rightarrow Y$ is étale, then*

$$\dim_k H_{dR}^1(X)^G = 2g_Y - \dim_k H^1(G, k) + \dim_k H^2(G, k).$$

In particular, if $G \cong \mathbb{Z}/p^n$ then $\dim_k H_{dR}^1(X)^G = 2g_Y$.

Proof. Let $\mathbb{H}^i(Y, \mathcal{F}^\bullet)$ be the i th hypercohomology of a complex $\mathcal{F}^\bullet$. Write also $\mathcal{H}^i(G, -)$ for the i th derived functor of the functor

$$\mathcal{F} \mapsto \mathcal{F}^G.$$

Since $X \rightarrow Y$ is étale, $\mathcal{H}^i(G, \pi_* \mathcal{F}) = 0$ for any $i > 0$ and any coherent sheaf $\mathcal{F}$ on X by [12, Proposition 2.1]. Therefore the spectral sequence [12, (3.4)] applied for the complex $\mathcal{F}^\bullet := \pi_* \Omega_{X/k}^\bullet$ yields $\mathbb{R}^i \Gamma^G(\pi_* \Omega_{X/k}^\bullet) = \mathbb{H}^i(Y, \pi_*^G \Omega_{X/k}^\bullet) = H_{dR}^1(Y)$, since $\pi_*^G \Omega_X^\bullet \cong \Omega_Y^\bullet$ (cf. [12, Corollary 2.4]). On the other hand, the seven-term exact sequence applied for the spectral sequence [12, (3.5)] yields:

$$0 \rightarrow H^1(G, H_{dR}^0(X)) \rightarrow H_{dR}^1(Y) \rightarrow H_{dR}^1(X)^G \rightarrow H^2(G, H_{dR}^0(X)) \rightarrow K,$$

where (since $H_{dR}^2(X) \cong k$ is a trivial $k[G]$ -module):

$$K := \ker(H_{dR}^2(Y) \rightarrow H_{dR}^2(X)^G) = \ker(k \xrightarrow{\text{id}} k) = 0.$$

Therefore, since $H_{dR}^0(X)^G \cong k$:

$$\begin{aligned} \dim_k H_{dR}^1(X)^G &= \dim_k H_{dR}^1(Y) - \dim_k H^1(G, k) + \dim_k H^2(G, k) \\ &= 2g_Y - \dim_k H^1(G, k) + \dim_k H^2(G, k). \end{aligned}$$

Finally, note that if G is cyclic then $\dim_k H^1(G, k) = \dim_k H^2(G, k)$ by [31, Theorem 6.2.2]. $\square$

Remark 4.4. The equality $\dim_k H^1(G, k) = \dim_k H^2(G, k)$ does not hold for non-cyclic groups. For example it is known [2, Corollary II.4.3, Theorem II.4.4] that the cohomological ring for the elementary abelian group $G = (\mathbb{Z}/p)^s$ is given by

$$H^*(G, \mathbb{F}_p) = \begin{cases} \mathbb{F}_2[x_1, \dots, x_s] & \text{if } p = 2 \\ \wedge(x_1, \dots, x_s) \otimes \mathbb{F}_p[x_1, \dots, x_s] & \text{if } p > 2 \end{cases}$$

Therefore, for $s > 1$ the degree one and two parts of the cohomological ring, which correspond to the first and second cohomology groups, have different dimensions.

Lemma 4.5. *Suppose that G is a p -group. If the G -cover $X \rightarrow Y$ has no étale subcovers, then the map*

$$\text{tr}_{X/Y} : H_{dR}^1(X) \rightarrow H_{dR}^1(Y)$$

is an epimorphism.

Proof. By induction, it suffices to prove this in the case when $G = \mathbb{Z}/p$. Consider the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, \Omega_X) & \longrightarrow & H^1_{dR}(X) & \longrightarrow & H^1(X, \mathcal{O}_X) \longrightarrow 0 \\ & & \downarrow \text{tr}_{X/Y} & & \downarrow \text{tr}_{X/Y} & & \downarrow \text{tr}_{X/Y} \\ 0 & \longrightarrow & H^0(Y, \Omega_Y) & \longrightarrow & H^1_{dR}(Y) & \longrightarrow & H^1(Y, \mathcal{O}_Y) \longrightarrow 0 \end{array} \quad (4.2)$$

where the rows are Hodge–de Rham exact sequences (cf. (2.2)). Recall that by [29, Theorem 1], in this case $H^0(X, \Omega_X)$ contains a copy of $k[G]^{\oplus g_Y}$ as a direct summand. Thus, since the image of $k[G]^{\oplus g_Y}$ under the trace map is $k^{\oplus g_Y}$, the dimension of the image of

$$\text{tr}_{X/Y} : H^0(X, \Omega_X) \rightarrow H^0(Y, \Omega_Y) \quad (4.3)$$

is g_Y . Therefore the map (4.3) is surjective. Similarly, by Serre’s duality, $H^1(X, \mathcal{O}_X)$ also contains $k[G]^{\oplus g_Y}$ as a direct summand and hence the trace map

$$\text{tr}_{X/Y} : H^1(X, \mathcal{O}_X) \rightarrow H^1(Y, \mathcal{O}_Y)$$

is surjective. Therefore, since the outer vertical maps in the diagram (4.2) are surjective, the trace map on the de Rham cohomology must be surjective as well. $\square$

Lemma 4.6. Assume that $\phi : Y' \rightarrow Y$ is a $\mathbb{Z}/p$ -subcover of a G -cover $X \rightarrow Y$, where $G \cong \mathbb{Z}/p^n$. Then:

$$p \cdot \sum_{Q \in B_{X/Y}} (u_{X/Y, Q} - 1) = \sum_{Q' \in B_{X/Y'}} (u_{X/Y', Q'} - 1) + (p-1) \cdot \sum_{Q \in B_{Y'/Y}} (l_{Y'/Y, Q}^{(1)} - 1).$$

Proof. Pick a point $Q \in B_{X/Y}$. If $Q \notin B_{Y'/Y}$ then $u_{X/Y, Q} = u_{X/Y', Q'}$ for all p points $Q' \in Y'(k)$ in the preimage of Q and:

$$p \cdot (u_{X/Y, Q} - 1) = \sum_{Q' \in \phi^{-1}(Q)} (u_{X/Y', Q'} - 1). \quad (4.4)$$

Assume now that $Q \in B_{Y'/Y}$. Then there exists a unique point $Q' \in Y'(k)$ in the preimage of Q through $\phi : Y' \rightarrow Y$. Moreover, since G_Q is a subgroup of $G \cong \mathbb{Z}/p^n$, which surjects onto $\mathbb{Z}/p$, we must have $G_Q = G$, $m_{X/Y, Q} = n$ and $m_{X/Y', Q'} = n-1$. Recall also that by [27, Example on p. 76] there exist integers $i_{X/Y, P}^{(0)}, i_{X/Y, P}^{(1)}, \dots$ such that for every $t \geq 1$:

$$\begin{aligned} u_{X/Y, P}^{(t)} &= i_{X/Y, P}^{(0)} + i_{X/Y, P}^{(1)} + \cdots + i_{X/Y, P}^{(t-1)} \\ l_{X/Y, P}^{(t)} &= i_{X/Y, P}^{(0)} + i_{X/Y, P}^{(1)} \cdot p + \cdots + i_{X/Y, P}^{(t-1)} \cdot p^{t-1}. \end{aligned}$$

Observe that:

$$\begin{aligned} i_{X/Y',P}^{(0)} &= i_{X/Y,P}^{(0)} + i_{X/Y,P}^{(1)} \cdot p, \\ i_{X/Y',P}^{(t)} &= p \cdot i_{X/Y,P}^{(t+1)} \quad \text{for } t > 0. \end{aligned}$$

This implies that

$$p \cdot (u_{X/Y,Q} - 1) = (u_{X/Y',Q'} - 1) + (p - 1) \cdot (l_{X/Y,Q}^{(1)} - 1). \quad (4.5)$$

Indeed, using the above formulas:

$$\begin{aligned} p \cdot (u_{X/Y,Q} - 1) &= p \cdot (i_{X/Y,Q}^{(0)} + \cdots + i_{X/Y,Q}^{(m_Q-1)} - 1) \\ &= (p - 1) \cdot (i_{X/Y,Q}^{(0)} - 1) + (i_{X/Y,Q}^{(0)} + p \cdot i_{X/Y,Q}^{(1)}) \\ &\quad + p \cdot (i_{X/Y,Q}^{(2)} + i_{X/Y,Q}^{(3)} + \cdots + i_{X/Y,Q}^{(m_Q-1)}) - 1 \\ &= (p - 1) \cdot (l_{X/Y,Q}^{(1)} - 1) + (i_{X/Y',Q'}^{(0)} + i_{X/Y',Q'}^{(1)} + \cdots - 1) \\ &= (p - 1) \cdot (l_{X/Y,Q}^{(1)} - 1) + (u_{X/Y',Q'} - 1). \end{aligned}$$

The proof follows by summing (4.4) and (4.5) over all $Q \in B_{X/Y}$. $\square$

Proof of Theorem 4.1. We proceed by induction on n . We define H' , H'' , $T^i M$, J_i and $\mathcal{J}_i$ as in Section 3. Also, we write $Y' := X/H'$ and $X'' := X/\langle \sigma^{p^{n-1}} \rangle$, see the diagram below.

$$\begin{array}{ccccc} & & X & & \\ \langle \sigma^p \rangle = H' \swarrow & & \downarrow \pi & \searrow \langle \sigma^{p^{n-1}} \rangle & \\ Y' & & Y & & X'' \\ & \searrow \phi & & \swarrow H'' & \end{array}$$

Note that H'' naturally acts on X'' and $X''/H'' \cong Y$. Let also $\mathcal{M} := H_{dR}^1(X)$ and $\mathcal{M}'' := H_{dR}^1(X'')$. Write $\mathcal{M}_0$ for the module (4.1). Note that the trace map $\text{tr}_{\langle \sigma^{p^{n-1}} \rangle} : \mathcal{M} \rightarrow \mathcal{M}$ equals the composition of $\text{tr}_{X/X''} : H_{dR}^1(X) \rightarrow H_{dR}^1(X'')$ with the inclusion of $H_{dR}^1(X'') \hookrightarrow H_{dR}^1(X)$. Moreover, by Lemma 4.5 if $X \rightarrow X''$ isn't étale then $\text{tr}_{X/X''} : H_{dR}^1(X) \rightarrow H_{dR}^1(X'')$ is surjective. Hence, by Lemma 3.2 (4) $\text{tr}_{X/X''}$ yields a k -linear isomorphism:

$$T^{j+p^n-p^{n-1}} \mathcal{M} \cong T_{H''}^j \mathcal{M}''. \quad (4.6)$$

The proof of Theorem 4.1 is divided into three cases.

Case I: the cover $X \rightarrow Y$ is étale.

By Lemma 4.3 and [12, Corollary 2.4] we have $\dim_k H_{dR}^1(X)^H = 2g_Y = \dim_k H^0(X, \Omega_X)^H + \dim_k H^1(X, \mathcal{O}_X)^H$. Therefore the Hodge–de Rham exact sequence splits by [12, Lemma 5.3] and

$$\begin{aligned} H_{dR}^1(X) &\cong H^0(X, \Omega_X) \oplus H^1(X, \mathcal{O}_X) \\ &\cong H^0(X, \Omega_X) \oplus H^0(X, \Omega_X)^\vee \end{aligned}$$

(the last isomorphism follows from Serre’s duality [17, Corollary III.7.7]). Now it suffices to note that $H^0(X, \Omega_X) \cong J_p^{g_Y-1} \oplus k$ by [28] (see also [21] and [29, Theorem 1]).

Case II: $n = 1$, the cover $X \rightarrow Y$ is not étale.

Note that $\mathcal{M}_0 = J_p^{\oplus 2g_Y} \oplus J_{p-1}^{\oplus \alpha}$, where $\alpha := \sum_{Q \in B_{X/Y}} (u_Q + 1) - 2$. Write

$$\mathcal{M} \cong \bigoplus_{i=1}^p J_i^{\oplus a_i}.$$

Observe that

$$\sum_{i=1}^p a_i = \dim_k \mathcal{M}^H = 2g_Y + \alpha. \quad (4.7)$$

Indeed, [12, Theorem 1.2] yields the following formula:

$$\begin{aligned} \delta &:= \dim_k H_{dR}^1(X)^H - \dim_k H^0(X, \Omega_X)^H - \dim_k H^1(X, \mathcal{O}_X)^H \\ &= \sum_{P \in X(k)} \left(u_P - 1 - 2 \left\lfloor \frac{u_P}{p} \right\rfloor \right) = \sum_{Q \in Y(k)} \left(u_Q - 1 - 2 \left\lfloor \frac{u_Q}{p} \right\rfloor \right) \end{aligned}$$

(for the last equality note that the summands are non-zero only for ramified points and that π yields a bijection between $\pi^{-1}(B_{X/Y})$ and $B_{X/Y}$). Put:

$$R' := \sum_{Q \in B_{X/Y}} \left\lfloor \frac{(u_Q + 1) \cdot (p - 1)}{p} \right\rfloor \cdot Q \in \text{Div}(Y). \quad (4.8)$$

Then, by [12, Corollary 2.4] and by Riemann–Roch theorem we have

$$\dim_k H^0(X, \Omega_X)^H = \dim_k H^0(Y, \Omega_Y(R')) = g_Y - 1 + \deg R'. \quad (4.9)$$

Moreover, for any $k[H]$ -module we have $\dim_k V^H = \dim_k V_H$ (one checks this easily by analyzing V as a direct sum of Jordan blocks) and $(V^H)^\vee \cong (V^\vee)_H$. Hence:

$$\begin{aligned}\dim_k H^1(X, \mathcal{O}_X)^H &= \dim_k (H^1(X, \mathcal{O}_X)^H)^\vee \\ &= \dim_k H^0(X, \Omega_X)_H = \dim_k H^0(X, \Omega_X)^H.\end{aligned}$$

This yields:

$$\begin{aligned}\dim_k H_{dR}^1(X)^H &= \delta + \dim_k H^0(X, \Omega_X)^H + \dim_k H^1(X, \mathcal{O}_X)^H \\ &= \sum_{Q \in Y(k)} \left(u_Q - 1 - 2 \left\lceil \frac{u_Q}{p} \right\rceil \right) + 2 \cdot (g_Y - 1 + \deg R') \\ &= 2g_Y + \alpha,\end{aligned}$$

which proves (4.7). Moreover, by (4.6) $T^p \mathcal{M} \cong \mathcal{M}'' \cong H_{dR}^1(Y)$, i.e. $a_p = 2g_Y$. By the formula for the exponent of the different in terms of the ramification groups (cf. [27, Proposition IV.§1.4]), one sees that the degree of the ramification divisor of $X \rightarrow Y$ equals $\sum_{Q \in B_{X/Y}} (u_Q + 1) \cdot (p - 1) = (\alpha + 2) \cdot (p - 1)$. Thus, using the Riemann–Hurwitz formula (cf. [17, Corollary IV.2.4]) we obtain:

$$\sum_{i=1}^p i \cdot a_i = \dim_k \mathcal{M} = 2g_X = 2g_Y \cdot p + \alpha \cdot (p - 1).$$

Hence:

$$\begin{aligned}\sum_{i=1}^{p-1} (p - 1 - i) \cdot a_i &= (p - 1) \sum_{i=1}^p a_i - \sum_{i=1}^p i \cdot a_i + a_p \\ &= (p - 1) \cdot (2g_Y + \alpha) - (2g_Y \cdot p + \alpha \cdot (p - 1)) + 2g_Y \\ &= 0.\end{aligned}$$

However, $p - 1 - i \geq 0$ with equality only for $i = p - 1$, which implies that $a_1 = \dots = a_{p-2} = 0$ and $a_{p-1} = \alpha$. This ends the proof in this case.

Case III: $n > 1$, the cover $X \rightarrow Y$ is not étale.

The assumption implies that $X \rightarrow X''$ is also not étale, since if σ^{p^j} fixes a point for some $1 \leq j < n$ then this point is also fixed by $\sigma^{p^{n-1}} = (\sigma^{p^j})^{p^{n-1-j}}$. We now show that $\dim_k T^i \mathcal{M} = \dim_k T^i \mathcal{M}_0$ separately for the cases $i \leq p^n - p^{n-1}$ and $i > p^n - p^{n-1}$. By induction hypothesis for H' acting on X , we have the following isomorphism of $k[H']$ -modules:

$$\begin{aligned}\mathcal{M}_{H'} &\cong \mathcal{J}_{p^{n-1}}^{2(g_{Y'}-1)} \oplus \mathcal{J}_{p^{n-1}-p^{n-1}-m'+1}^2 \oplus \bigoplus_{\substack{Q \in Y'(k) \\ Q \neq Q_1}} \mathcal{J}_{p^{n-1}-p^{n-1}/e'_Q}^2 \\ &\quad \oplus \bigoplus_{Q \in Y'(k)} \bigoplus_{t=0}^{m'_Q-1} \mathcal{J}_{p^{n-1}-p^{n-1}+t/e'_Q}^{u_{X/Y',Q}^{(t+1)} - u_{X/Y',Q}^{(t)}}\end{aligned}\tag{4.10}$$

where $e'_Q := e_{X/Y',Q}$, $m'_Q := m_{X/Y',Q}$, $m' := m_{X/Y'}$ and $Q_1 \in \pi^{-1}(Q_0)$. Note also that

$$\begin{aligned} \#B_{X/Y'} &= p \cdot \#\{Q \in B_{X/Y} : e_{Y'/Y,Q} = 1\} + \#\{Q \in B_{X/Y} : e_{Y'/Y,Q} = p\} \\ &= p \cdot (\#B - \#B_{Y'/Y}) + \#B_{Y'/Y} \\ &= p \cdot \#B - (p-1) \cdot \#B_{Y'/Y}. \end{aligned} \quad (4.11)$$

Therefore, for $i \leq p^{n-1} - p^{n-2}$, using induction hypothesis (cf. (4.10)):

$$\dim_k T_{H'}^i \mathcal{M} = 2(g_{Y'} - 1) + 2 + 2(\#B_{X/Y'} - 1) + \sum_{Q' \in Y'(k)} (u_{X/Y',Q'} - 1)$$

(by the Riemann–Hurwitz formula, cf. [17, Corollary IV.2.4] and the formula for the exponent of the different from [27, Proposition IV.§1.4])

$$\begin{aligned} &= 2p(g_Y - 1) + \sum_{Q' \in Y'(k)} (p-1) \cdot (l_{Y'/Y,Q'}^{(1)} + 1) \\ &+ 2 + 2(\#B_{X/Y'} - 1) + \sum_{Q' \in Y'(k)} (u_{X/Y',Q'} - 1) \\ &= 2p(g_Y - 1) + \sum_{Q' \in Y'(k)} (p-1) \cdot (l_{Y'/Y,Q'}^{(1)} - 1) \\ &+ 2 \cdot (p-1) \cdot \#B_{Y'/Y} + 2 + 2(\#B_{X/Y'} - 1) + \sum_{Q' \in Y'(k)} (u_{X/Y',Q'} - 1) \end{aligned}$$

(by (4.11))

$$\begin{aligned} &= 2p(g_Y - 1) + \sum_{Q' \in Y'(k)} (p-1) \cdot (l_{Y'/Y,Q'}^{(1)} - 1) + 2 \cdot p \cdot \#B \\ &+ \sum_{Q' \in Y'(k)} (u_{X/Y',Q'} - 1) \end{aligned}$$

(using Lemma 4.6)

$$= p \cdot \left(2(g_Y - 1) + 2\#B + \sum_{Q \in Y(k)} (u_{X/Y,Q} - 1) \right).$$

In particular, $\dim_k T_{H'}^1 \mathcal{M} = \dots = \dim_k T_{H'}^{p^{n-1}-p^{n-2}} \mathcal{M}$, which implies by Lemma 3.2 (3) that

$$\dim_k T^1 \mathcal{M} = \dots = \dim_k T^{p^n-p^{n-1}} \mathcal{M}.$$

Noting that (cf. Lemma 3.2 (2)):

$$\dim_k T_{H'}^1 \mathcal{M} = \dim_k T^1 \mathcal{M} + \dots + \dim_k T^p \mathcal{M},$$

we obtain that for any $1 \leq i \leq p^n - p^{n-1}$:

$$\begin{aligned} \dim_k T^i \mathcal{M} &= \frac{1}{p} \dim_k T_{H'}^1 \mathcal{M} \\ &= 2(g_Y - 1) + 2 + 2(\#B - 1) + \sum_{Q \in Y(k)} (u_{X/Y, Q} - 1) \\ &= \dim_k T^i \mathcal{M}_0. \end{aligned}$$

Now we treat the case when $i > p^n - p^{n-1}$. Let $i = p^n - p^{n-1} + j$ and let N be chosen so that $j \in (p^{n-1} - p^{N+1}, p^{n-1} - p^N]$. Then:

$$\begin{aligned} \dim_k T^j \mathcal{M}_0 &= 2 \cdot (g_Y - 1) + 2 \cdot \llbracket N \geq n - m \rrbracket \\ &\quad + 2 \cdot \#\{Q \in Y(k) \setminus \{Q_0\} : N \geq n - m_Q\} \\ &\quad + \sum_{Q \in Y(k)} \sum_{t=0}^{m_Q-1} \llbracket N \geq n + t - m_Q \rrbracket \cdot (u_Q^{(t+1)} - u_Q^{(t)}). \end{aligned}$$

On the other hand, by (4.6) and by induction assumption for $X'' \rightarrow Y$:

$$\begin{aligned} \dim_k T^i \mathcal{M} &= \dim_k T_{H''}^j \mathcal{M}'' = 2 \cdot (g_Y - 1) + 2 \cdot \llbracket N \geq (n-1) - (m-1) \rrbracket \\ &\quad + 2 \cdot \#\{Q \in Y(k) \setminus \{Q_0\} : N \geq (n-1) - m_{X''/Y, Q}\} \\ &\quad + \sum_{Q \in Y(k)} \sum_{t=0}^{m_{X''/Y, Q}} \llbracket N \geq (n-1) + t - m_{X''/Y, Q} \rrbracket \cdot (u_{X''/Y, Q}^{(t+1)} - u_{X''/Y, Q}^{(t)}) \\ &= 2 \cdot (g_Y - 1) + 2 \cdot \llbracket N \geq n - m \rrbracket \\ &\quad + 2 \cdot \#\{Q \in Y(k) \setminus \{Q_0\} : N \geq n - m_Q\} \\ &\quad + \sum_{Q \in Y(k)} \sum_{t=0}^{m_Q-1} \llbracket N \geq n + t - m_Q \rrbracket \cdot (u_Q^{(t+1)} - u_Q^{(t)}) \\ &= \dim_k T^i \mathcal{M}_0. \end{aligned}$$

This ends the proof. $\square$

5. Proof of main theorem

In this section we prove Main Theorem, assuming a formula for the cohomology of a cover with a p -Sylow subgroup of order p . We prove this formula in the next section (cf. Theorem 6.1). The following result allows us to reduce the problem to the case when G is of the form (3.1) with $C = \mathbb{Z}/c$, $p \nmid c$ and k is an algebraically closed field.

Lemma 5.1. *Let G and k be as in Main Theorem. Suppose M is a finitely generated $k[G]$ -module.*

- (1) *The $k[G]$ -module structure of M is uniquely determined by the restrictions $M|_H$ as H ranges over all subgroups of G of the form (3.1) with $C \cong \mathbb{Z}/c$, $p \nmid c$.*
- (2) *The $k[G]$ -module structure of M is uniquely determined by the $\bar{k}[G]$ -module structure of $M \otimes_k \bar{k}$.*

Proof. (1) This follows easily from Conlon induction theorem (cf. [9, Theorem (80.51)]), see e.g. [5, Lemma 3.2].

(2) This is [5, Proposition 3.5. (iii)]. $\square$

The following simple lemma will be used in the proof of Main Theorem.

Lemma 5.2. *Keep the above notation. Let M, N be $k[C]$ -modules. If*

$$N \cong M \oplus M^\chi \oplus \cdots \oplus M^{\chi^{p-1}},$$

then M is uniquely determined by N .

Proof. Note that

$$N \cong M^{\oplus 2} \oplus M^\chi \oplus M^{\chi^2} \oplus \cdots \oplus M^{\chi^{p-2}}.$$

By tensoring this isomorphism by χ^i we obtain:

$$N^{\chi^i} \cong (M^{\chi^i})^{\oplus 2} \oplus M^{\chi^{i+1}} \oplus M^{\chi^{i+2}} \oplus \cdots \oplus M^{\chi^{i+p-2}} \cong (M^{\chi^i})^{\oplus 2} \oplus \bigoplus_{\substack{j=0 \\ j \neq i}}^{p-2} M^{\chi^j}$$

for $i = 0, \dots, p-2$. Therefore:

$$M^{\oplus p} \oplus N^\chi \oplus N^{\chi^2} \oplus \cdots \oplus N^{\chi^{p-2}} \cong N^{\oplus (p-1)}. \quad (5.1)$$

Indeed, for the proof of (5.1) note that

$$\begin{aligned} M^{\oplus p} \oplus N^\chi \oplus N^{\chi^2} \oplus \cdots \oplus N^{\chi^{p-2}} &\cong M^{\oplus p} \oplus \bigoplus_{i=1}^{p-2} \left((M^{\chi^i})^{\oplus 2} \oplus \bigoplus_{\substack{j=0 \\ j \neq i}}^{p-2} M^{\chi^j} \right) \\ &\cong \left(M^{\oplus 2} \oplus M^\chi \oplus M^{\chi^2} \oplus \cdots \oplus M^{\chi^{p-2}} \right)^{\oplus (p-1)} \cong N^{\oplus (p-1)}. \end{aligned}$$

The isomorphism (5.1) clearly proves the thesis. $\square$

Proof of Main Theorem. As explained at the beginning of this section, it suffices to show this in the case when G is of the form (3.1) and $k = \bar{k}$ by Lemma 5.1. Let $Y := X/H$. Similarly as in the proof of Theorem 4.1, we write $H' := \langle \sigma^p \rangle \cong \mathbb{Z}/p^{n-1}$, $H'' := H/\langle \sigma^{p^{n-1}} \rangle \cong \mathbb{Z}/p^{n-1}$, $Y' := X/H'$ and $X'' := X/\langle \sigma^{p^{n-1}} \rangle$. Observe that the ramification data of the covers $X'' \rightarrow Y$ and $X \rightarrow Y'$ depends only on the ramification data of $X \rightarrow Y$.

We prove the result by induction on n . For $n = 0$ this follows by Chevalley–Weil theorem. The rest of the proof is again divided into three cases.

Case I: the cover $X \rightarrow Y$ is étale.

In this case we deduce as in the proof of Theorem 4.1 that

$$H_{dR}^1(X) \cong H^0(X, \Omega_X) \oplus H^0(X, \Omega_X)^\vee$$

as $k[G]$ -modules and hence the result follows by [5, Theorem 1.1].

Case II: $n = 1$, the cover $X \rightarrow Y$ is not étale.

This case will be addressed in the next section (see Theorem 6.1).

Case III: $n > 1$, the cover $X \rightarrow Y$ is not étale.

Recall that by Lemma 3.2 (5) the isomorphism class of $\mathcal{M}$ is uniquely determined by the $k[C]$ -modules $T^1\mathcal{M}, \dots, T^{p^n}\mathcal{M}$. Lemma 3.2 (3) and Theorem 4.1 yield an isomorphism of $k[C]$ -modules:

$$T^{i+1}\mathcal{M} \cong (T^1\mathcal{M})^{\chi^{-i}} \quad (5.2)$$

for $i < p^n - p^{n-1}$. By Lemma 3.2 (2), for $i \leq p^{n-1} - p^{n-2}$:

$$\begin{aligned} T_{H'}^i\mathcal{M} &\cong T^{pi-p+1}\mathcal{M} \oplus \dots \oplus T^{pi}\mathcal{M} \\ &\cong (T^1\mathcal{M})^{\chi^{-(pi-p)}} \oplus (T^1\mathcal{M})^{\chi^{-(pi-p+1)}} \oplus \dots \oplus (T^1\mathcal{M})^{\chi^{-(pi-1)}}. \end{aligned}$$

Take $i = 1$. By induction assumption, the $k[C]$ -module structure of $T_{H'}^1\mathcal{M}$ is uniquely determined by the ramification data. Thus, by Lemma 5.2 for $N := T^1\mathcal{M}$ and by (5.2) the $k[C]$ -structure of the modules $T^i\mathcal{M}$ is uniquely determined by the ramification data for $i \leq p^n - p^{n-1}$. By Lemma 3.2 (4) and (4.6), $\mathrm{tr}_{X/X''}$ yields an isomorphism:

$$T^{i+p^n-p^{n-1}}\mathcal{M} \cong T_{H''}^i\mathcal{M}'' \quad (5.3)$$

Thus, by induction hypothesis for $\mathcal{M}''$, the $k[C]$ -structure of $T^{i+p^n-p^{n-1}}\mathcal{M}$ is determined by the ramification data as well. $\square$

6. Covers with Sylow subgroup of order p

In this section we assume that G is of the form (3.1) with $n = 1$. Assume that X is a smooth projective curve with an action of G and let $\pi : X \rightarrow Y := X/H$. The goal of this section is to prove the following variant of Chevalley–Weil formula in this context.

Theorem 6.1. *Keep the above notation. Assume that k is algebraically closed. If G acts on a curve X and the cover $X \rightarrow Y$ is not étale, then:*

$$H_{dR}^1(X) \cong J_p(V_1) \oplus J_{p-1}(V_2),$$

where the $k[C]$ -modules V_1 and V_2 are determined by the isomorphisms:

$$V_1 \cong H^0(Y, \Omega_Y) \oplus H^1(Y, \mathcal{O}_Y),$$

$$V_2 \cong H^0(Y, \Omega_Y(R')) \oplus H^1(Y, \mathcal{O}_Y(-D))^{x^{-1}} \oplus H^1(Y, \mathcal{O}_Y)^{x^{-1}} \oplus H^0(Y, \Omega_Y),$$

the divisor R' is given by (4.8) and $D \in \text{Div}(Y)$ is defined as:

$$D := \sum_{Q \in B_{X/Y}} \left\lceil \frac{u_{X/Y, Q}}{p} \right\rceil \cdot Q.$$

Here $W := W_1 \ominus W_2$ means that $W_1 \cong W \oplus W_2$ (note that if such a W exists, it is unique). By Proposition 2.1, the representations defining V_1 and V_2 are determined by the ramification data.

We sketch now briefly the idea behind the proof of Theorem 6.1. Using techniques similar as in the proof of Main Theorem, one concludes that the $k[G]$ -module structure of $H_{dR}^1(X)$ is determined by the $k[C]$ -structure of $T^1 H_{dR}^1(X)$ and $T^p H_{dR}^1(X)$. Moreover, $T^p H_{dR}^1(X) \cong H_{dR}^1(Y)$. In order to determine $H_{dR}^1(X)^H$ consider the sequence of invariants arising from the Hodge–de Rham exact sequence (2.2):

$$0 \rightarrow H^0(X, \Omega_X)^H \rightarrow H_{dR}^1(X)^H \rightarrow H^1(X, \mathcal{O}_X)^H. \quad (6.1)$$

We compute explicitly the image $\mathbb{I}_{dR}$ of $H_{dR}^1(X)^H$ in $H^1(X, \mathcal{O}_X)^H$ (cf. Proposition 6.6 (2)). To this end we need some auxiliary results. For a sheaf with an action of H we write again $\mathcal{F}^{(i)} := \ker((\sigma - 1)^i : \mathcal{F} \rightarrow \mathcal{F})$ and $T^i \mathcal{F} := \mathcal{F}^{(i)} / \mathcal{F}^{(i-1)}$. Since in the sequel we work with sheaves on Y , we abbreviate $\pi_* \mathcal{O}_X$ to $\mathcal{O}_Y$, $\pi_* \mathcal{O}_X^{(i)}$ to $\mathcal{O}_Y^{(i)}$ etc. Note that $H^i(X, \mathcal{F}) = H^i(Y, \pi_* \mathcal{F})$ for any quasicoherent $\mathcal{O}_X$ -module $\mathcal{F}$ and any $i \geq 0$, since π is an affine morphism. We say that a function $y \in k(X)$ is an *Artin–Schreier generator in standard form on U* , if $\sigma(y) = y + 1$ and the poles of $f := y^p - y \in k(Y)$ contained in U have order non-divisible by p . Note that for any affine open subset $U \subset Y$ there exists an Artin–Schreier generator in standard form on U . Moreover, if y is an Artin–Schreier generator, then any other generator is of the form $y + g$ for $g \in \mathcal{O}_Y(U)$.

Lemma 6.2. *Assume that y is an Artin–Schreier generator in standard form on an open subset $U \subset Y$. Then:*

$$\mathcal{O}_X|_U = \bigoplus_{i=0}^{p-1} y^i \mathcal{O}_U(-D_i),$$

where $D_i := \sum_{Q \in B_{X/Y}} \left\lceil \frac{u_{X/Y, Q} \cdot i}{p} \right\rceil \cdot Q \in \text{Div}(Y)$. In particular, for any $j = 0, \dots, p-1$ we have the following isomorphism of $\mathcal{O}_Y$ -modules:

$$T^j \mathcal{O}_X \cong \mathcal{O}_Y(-D_{j-1}). \quad (6.2)$$

Proof. By [13, (7.8)] for any $g = g_0 + g_1 \cdot y + \dots + g_{p-1} \cdot y^{p-1} \in k(X)$ and $P \in U$ we have $\text{ord}_P(g) = \min\{\text{ord}_P(g_i \cdot y^i) : i = 0, \dots, p-1\}$. Moreover, for $Q \in B_{X/Y}$ and $P \in \pi^{-1}(Q)$ we have $\text{ord}_P(y) = \frac{1}{p} \text{ord}_P(f) = \text{ord}_Q(f) = -u_{X/Y, P}$ (cf. [13, (7.2)] for the last equality). Therefore $\text{ord}_P(g) \geq 0$ holds if and only if $\text{ord}_Q(g_i) \geq \frac{u_{X/Y, P} \cdot i}{p}$. This proves the first equality. In order to find $T^j \mathcal{O}_X$, note that

$$\mathcal{O}_X^{(j)}|_U = \bigoplus_{i=0}^{j-1} y^i \mathcal{O}_U(-D_i). \quad (6.3)$$

This clearly implies that we have an isomorphism:

$$T^j \mathcal{O}_X|_U \cong \mathcal{O}_Y(-D_{j-1})|_U, \quad g = g_0 + y \cdot g_1 + \dots + y^{j-1} \cdot g_{j-1} \mapsto g_{j-1}.$$

One easily checks that this isomorphism does not depend on the choice of y . Hence, by picking a cover of Y by open affine subsets and choosing an Artin–Schreier generator in standard form on each of them, we obtain the desired isomorphism. $\square$

Lemma 6.3. *Keep the above notation. Assume that M is a $k[G]$ -module and $M = M^{(2)}$. Then $M_H \oplus M^\chi \cong (M^H)^\chi \oplus M$ as $k[C]$ -modules.*

Proof. By a similar reasoning as in Lemma 3.2 (1), $\sigma - 1$ is χ^{-1} -linear. This yields the following short exact sequences:

$$\begin{aligned} 0 \rightarrow M^H \rightarrow M \xrightarrow{\sigma-1} \text{im}(\sigma-1)^{\chi^{-1}} \rightarrow 0, \\ 0 \rightarrow \text{im}(\sigma-1) \rightarrow M \rightarrow M_H \rightarrow 0. \end{aligned}$$

Therefore, since the category of $k[C]$ -modules is semisimple:

$$\text{im}(\sigma-1) \cong M^\chi \oplus (M^H)^\chi \cong M \oplus M_H,$$

which yields $M_H \oplus M^\chi \cong (M^H)^\chi \oplus M$. $\square$

Lemma 6.4. *Keep the above notation. Then, as $\mathcal{O}_Y$ -modules with an action of C :*

$$((\mathcal{O}_X^{(2)})^\vee)^H \cong \mathcal{O}_Y(D)^\chi,$$

where D is defined as in Theorem 6.1.

Proof. The following maps are mutually inverse isomorphisms of $\mathcal{O}_Y$ -modules:

$$\Psi : \mathcal{O}_Y(D) \rightarrow \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{O}_X^{(2)}, \mathcal{O}_Y)^H, \quad \Psi(h) = \varphi_h,$$

$$\Phi : \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{O}_X^{(2)}, \mathcal{O}_Y)^H \rightarrow \mathcal{O}_Y(D), \quad \Phi(\varphi) = h_\varphi,$$

where for an open subset $U \subset Y$:

$$\varphi_h(g) := h \cdot (\sigma(g) - g) \quad \text{for } g \in H^0(U, \mathcal{O}_X^{(2)}) \text{ and } h \in H^0(U, \mathcal{O}_Y(D)),$$

$$h_\varphi := \frac{\varphi(g)}{\sigma(g) - g} \quad \text{for } \varphi \in \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{O}_X^{(2)}|_U, \mathcal{O}_U)^H$$

$$\text{and any } g \in \mathcal{O}_X^{(2)}(U) \setminus \mathcal{O}_Y(U).$$

In order to check that those maps are well-defined, pick an arbitrary affine open set $U \subset Y$ and an Artin–Schreier generator y in standard form on U . Then, by (6.3) we have $(\mathcal{O}_X^{(2)})|_U = \mathcal{O}_U \oplus y\mathcal{O}_U(-D)$. Hence, if $g = g_0 + g_1 \cdot y \in H^0(U, \mathcal{O}_X^{(2)})$ and $h \in H^0(U, \mathcal{O}_Y(D))$, then $g_0 \in H^0(U, \mathcal{O}_Y)$, $g_1 \in H^0(U, \mathcal{O}_Y(-D))$ and $\varphi_h(g) = g_1 \cdot h \in H^0(U, \mathcal{O}_Y)$. One easily checks that φ_h is H -invariant.

We check now that if $\varphi \in H^0(U, \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{O}_X^{(2)}, \mathcal{O}_Y)^H)$ then h_φ is well-defined. Firstly, h_φ does not depend on the choice of g . Indeed, if $g' = g'_0 + y \cdot g'_1$ then:

$$\frac{\varphi(g)}{\sigma(g) - g} - \frac{\varphi(g')}{\sigma(g') - g'} = \frac{\varphi(g_0g'_1 - g'_0g_1)}{g_1g'_1} = \frac{\varphi((\sigma - 1) \cdot g'')}{g_1g'_1} = 0,$$

where $g'' := (g_0g'_1 - g'_0g_1) \cdot y \in \mathcal{O}_X^{(2)}$, since $g_0g'_1 - g'_0g_1 \in H^0(U, \mathcal{O}_Y(-D))$. Secondly, pick $g_1 \in k(Y)$ in such a way that $\mathrm{div}(g_1|_U)$ (the divisor of g_1 restricted to U) equals $D|_U$ (this is possible, since U is affine). Then for $g := y \cdot g_1$ we clearly have $h_\varphi = \frac{\varphi(g)}{g_1} \in H^0(U, \mathcal{O}_Y(D))$. One easily checks that Ψ and Φ are inverse to each other. Finally, we observe that when accounting for the action of C , the morphism Ψ is χ^{-1} -linear, which ends the proof. $\square$

Lemma 6.5. *As a $k[C]$ -module:*

$$H^0(Y, (\mathcal{O}_X^{(2)})^\vee \otimes \Omega_Y) \cong H^0(Y, \Omega_Y) \oplus H^0(Y, \Omega_Y(D))^\chi.$$

Proof. Consider the following exact sequence:

$$0 \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_X^{(2)} \rightarrow T^2\mathcal{O}_X^{(2)} \rightarrow 0.$$

Note that $T^2\mathcal{O}_X^{(2)} \cong T^2\mathcal{O}_X \cong \mathcal{O}_Y(-D)$ as sheaves on Y by (6.2). Moreover, since the isomorphism is χ^{-1} -linear, $T^2\mathcal{O}_X^{(2)} \cong \mathcal{O}_Y(-D)^{\chi^{-1}}$ as sheaves with an action of C . Therefore, by considering the associated long exact sequence and noting that $H^0(Y, \mathcal{O}_Y(-D)) = 0$:

$$0 \rightarrow H^1(Y, \mathcal{O}_Y) \rightarrow H^1(Y, \mathcal{O}_X^{(2)}) \rightarrow H^1(Y, \mathcal{O}_Y(-D))^{\chi^{-1}} \rightarrow 0.$$

Using Serre's duality and the fact that the category of $k[C]$ -modules is semi-simple, we obtain

$$H^0(Y, (\mathcal{O}_X^{(2)})^\vee \otimes \Omega_Y) \cong H^1(Y, \mathcal{O}_X^{(2)})^\vee \cong H^0(Y, \Omega_Y) \oplus H^0(Y, \Omega_Y(D))^\chi. \quad \square$$

Proposition 6.6.

(1) *There exists an isomorphism of $k[C]$ -modules:*

$$H^1(X, \mathcal{O}_X^{(2)})^H \cong H^1(Y, \mathcal{O}_Y) \oplus H^1(Y, \mathcal{O}_Y(-D))^{\chi^{-1}} \ominus H^1(Y, \mathcal{O}_Y)^{\chi^{-1}}.$$

(2) *The module $\mathbb{I}_{dR}$ equals $\mathbb{I} := \text{im} \left(H^1(X, \mathcal{O}_X^{(2)})^H \rightarrow H^1(X, \mathcal{O}_X)^H \right)$.*

Proof. (1) Write $M := H^0(Y, (\mathcal{O}_X^{(2)})^\vee \otimes \Omega_Y)$. Then, by Serre's duality:

$$H^1(X, \mathcal{O}_X^{(2)})^H \cong (M_H)^\vee$$

Moreover, by Lemma 6.4:

$$\begin{aligned} M^H &= H^0(Y, (\mathcal{O}_X^{(2)})^\vee \otimes \Omega_Y)^H \\ &\cong H^0(Y, ((\mathcal{O}_X^{(2)})^\vee)^H \otimes \Omega_Y) \\ &\cong H^0(Y, \Omega_Y(D))^\chi. \end{aligned} \tag{6.4}$$

Therefore, using Lemma 6.3, Lemma 6.5 and (6.4):

$$\begin{aligned} M_H &\cong (H^0(Y, \Omega_Y) \oplus H^0(Y, \Omega_Y(D))^\chi) \oplus H^0(Y, \Omega_Y(D))^{\chi^2} \\ &\ominus (H^0(Y, \Omega_Y) \oplus H^0(Y, \Omega_Y(D))^\chi)^\chi \\ &\cong H^0(Y, \Omega_Y) \oplus H^0(Y, \Omega_Y(D))^\chi \ominus H^0(Y, \Omega_Y)^\chi. \end{aligned}$$

This, along with Serre's duality, finishes the proof.

(2) Recall that as a $k[H]$ -module $H_{dR}^1(X)$ is a direct sum of copies of J_p and J_{p-1} by Theorem 4.1. This easily implies that

$$H_{dR}^1(X)^H = \text{im}((\sigma - 1)^{p-2} : H_{dR}^1(X)^{(p-1)} \rightarrow H_{dR}^1(X)^{(p-1)}).$$

Therefore the map $H_{dR}^1(X)^H \rightarrow H^1(X, \mathcal{O}_X)^H$ factors through the group

$$L := \text{im}((\sigma - 1)^{p-2} : H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X)).$$

Let $\mathcal{I}$ be the sheaf image of $(\sigma - 1)^{p-2} : \mathcal{O}_X \rightarrow \mathcal{O}_X$. Since $(\sigma - 1)^{p-2} : \mathcal{O}_X \rightarrow \mathcal{O}_X$ factors through $\mathcal{I}$, the map $L \rightarrow H^1(X, \mathcal{O}_X)$ factors through $H^1(X, \mathcal{I})$. On the other hand, the inclusion $\mathcal{I} \subset \mathcal{O}_X^{(2)}$ implies that $H^1(X, \mathcal{I}) \rightarrow H^1(X, \mathcal{O}_X)$ factors through $H^1(X, \mathcal{O}_X^{(2)})$. Hence $\mathbb{I}_{dR} \subset \mathbb{I}$. Note that the map $H^1(X, \mathcal{O}_X^{(2)}) \rightarrow H^1(X, \mathcal{O}_X)$ is injective. Indeed, using (6.2) and induction one may easily prove that $H^0(X, \mathcal{O}_X/\mathcal{O}_X^{(j)}) = 0$ for any $0 \leq j \leq p-1$. Hence:

$$\ker \left(H^1(X, \mathcal{O}_X^{(2)}) \rightarrow H^1(X, \mathcal{O}_X) \right) = H^0(X, \mathcal{O}_X/\mathcal{O}_X^{(2)}) = 0.$$

Recall that $\dim_k H_{dR}^1(X)^H = 2g_Y + \alpha$ (cf. (4.7)) and $\dim_k H^0(X, \Omega_X)^H = g_Y - 1 + \deg R'$ (cf. (4.9)). Moreover, the identity:

$$(u+1) - \left\lfloor \frac{(u+1) \cdot (p-1)}{p} \right\rfloor = \left\lceil \frac{u}{p} \right\rceil$$

valid for any $u \in \mathbb{Z}$, $p \nmid u$ yields $\alpha - \deg R' = \deg D - 2$. Therefore, by part (1):

$$\begin{aligned} \dim_k \mathbb{I}_{dR} &= \dim_k H_{dR}^1(X)^H - \dim_k H^0(X, \Omega_X)^H \\ &= g_Y - 1 + \deg D = \dim_k H^0(Y, \Omega_Y(D)) = \dim_k \mathbb{I}. \end{aligned}$$

This shows that $\mathbb{I}_{dR} = \mathbb{I}$ and finishes the proof. $\square$

Proof of Theorem 6.1. By Theorem 4.1 there exist $k[C]$ -modules V_1, V_2 such that:

$$H_{dR}^1(X) \cong J_p(V_1) \oplus J_{p-1}(V_2).$$

By the definition of $\mathbb{I}_{dR}$:

$$0 \rightarrow H^0(X, \Omega_X)^H \rightarrow H_{dR}^1(X)^H \rightarrow \mathbb{I}_{dR} \rightarrow 0.$$

Recall that $\pi_*^H \Omega_X \cong \Omega_Y(R')$ (cf. [12, Corollary 2.4]). Therefore, by Proposition 6.6, we have the following isomorphism of $k[C]$ -modules:

$$\begin{aligned} H_{dR}^1(X)^H &\cong H^0(X, \Omega_X)^H \oplus \mathbb{I}_{dR} \cong H^0(Y, \pi_*^H \Omega_X) \oplus \mathbb{I}_{dR} \\ &\cong H^0(Y, \Omega_Y(R')) \oplus H^1(Y, \mathcal{O}_Y) \\ &\quad \oplus H^1(Y, \mathcal{O}_Y(-D))^{\chi^{-1}} \oplus H^1(Y, \mathcal{O}_Y)^{\chi^{-1}}. \end{aligned}$$

On the other hand, $H_{dR}^1(X)^H \cong V_1 \oplus V_2$, which yields a formula for $V_1 \oplus V_2$. Moreover, the map $\text{tr}_{X/Y} : H_{dR}^1(X) \rightarrow H_{dR}^1(Y)$ induces an isomorphism

$$T^p H_{dR}^1(X) \cong H_{dR}^1(Y) \cong H^0(Y, \Omega_Y) \oplus H^1(Y, \mathcal{O}_Y)$$

(see (5.3)). One finishes the proof, by noting that

$$T^p H_{dR}^1(X) \cong V_1^{\chi^{-p+1}} \cong V_1. \quad \square$$

7. An example – a superelliptic curve with a metacyclic action

Let $p > 2$ be a prime and let m be a natural number such that $p \nmid m$. Let k be an algebraically closed field of characteristic p . Fix a primitive root of unity $\zeta \in \overline{\mathbb{F}}_p^\times$ of order $m \cdot (p-1)$. Note that $\zeta^m \in \mathbb{F}_p$. Let $V \subset \overline{\mathbb{F}}_p$ be a $\mathbb{F}_p$ -linear space of order p^n containing $\mathbb{F}_p$. In this section we compute the equivariant structure of the de Rham cohomology for the superelliptic curve X with the affine part given by:

$$y^m = f_V(x) := \prod_{v \in V} (x - v).$$

Note that for $m = 2$, $V = \mathbb{F}_{p^2}$ this curve was considered e.g. in [6, Section 4]. It is a curve of genus $\frac{1}{2}(p^n - 1)(m - 1)$ (see e.g. [30, p. 226]) with an action of the group $G' := V \rtimes_{\chi'} C$, where $C := \langle \rho \rangle \cong \mathbb{Z}/(m \cdot (p-1))$ and

$$\chi' : C \rightarrow \text{Aut}(V), \quad \chi'(\rho)(v) = \zeta^{-m} \cdot v.$$

This action is given by:

$$\begin{aligned} \sigma_v(x, y) &= (x + v, y) \quad \text{for } v \in V, \\ \rho(x, y) &= (\zeta^{-m} \cdot x, \zeta^{-1} \cdot y) \end{aligned}$$

(note that G' acts on $X(k)$ from the left and on $k(X)$ from the right). In the sequel we restrict our attention to a subgroup of G' that satisfies the assumption of Main Theorem. Namely, let $G := H \rtimes_{\chi} C$, where $H := \mathbb{F}_p \subset V$ and (writing σ for the generator of H):

$$\chi : C \rightarrow \text{Aut}(H), \quad \chi(\rho)(\sigma) = \sigma^{\zeta^{-m}}.$$

Let $\psi : C \rightarrow k^\times$ be the primitive character determined by $\psi(\rho) = \zeta$. Note that χ might be identified with ψ^m .

Proposition 7.1. *Keep the above notation. There exists an isomorphism of $k[G]$ -modules:*

$$H_{dR}^1(X) \cong J_p(V_1) \oplus J_{p-1}(V_2),$$

where the $k[C]$ -modules V_1, V_2 are as follows:

$$V_1 \cong \bigoplus_{l=0}^{m \cdot (p-1)} (\psi^l)^{\oplus(\alpha_l + \alpha_{-l})}$$

$$V_2 \cong \bigoplus_{l=0}^{m \cdot (p-1)} (\psi^l)^{\oplus (\gamma_l + \beta_{-l-m} - \alpha_{-l+m} - \alpha_l)}$$

and:

$$\begin{aligned}\alpha_l &:= \delta_l + \left\langle \frac{-c_3 \cdot l}{m \cdot (p-1)} \right\rangle + \llbracket l = 0 \rrbracket, \\ \beta_l &:= \delta_l + \frac{\lceil m/p \rceil}{m \cdot (p-1)} + \left\langle \frac{-\lceil m/p \rceil - c_3 \cdot l}{m \cdot (p-1)} \right\rangle, \\ \gamma_l &:= \delta_l + \frac{\lfloor (m+1) \cdot (p-1)/p \rfloor}{m \cdot (p-1)} + \left\langle \frac{-\lfloor (m+1) \cdot (p-1)/p \rfloor - c_3 \cdot l}{m \cdot (p-1)} \right\rangle, \\ \delta_l &:= -1 + \left\langle \frac{l}{m} \right\rangle \cdot \frac{p^{n-1} - 1}{p-1} + \left\langle \frac{l}{m \cdot (p-1)} \right\rangle.\end{aligned}$$

Here c_1, c_2 are any integers such that $c_2 \cdot p^{n-1} - c_1 \cdot m = 1$ and c_3 is the multiplicative inverse of $c_2 - m \cdot c_1$ modulo $m \cdot (p-1)$.

Let Y be the superelliptic curve given by the equation:

$$y^m = f_{\overline{V}}(z) := \prod_{\overline{v} \in \overline{V}} (z - \overline{v}),$$

where $\overline{V} := \{v^p - v : v \in V\}$. Observe that the map $V \rightarrow \overline{V}$, $v \mapsto v^p - v$ yields an isomorphism $V/\mathbb{F}_p \cong \overline{V}$. The map

$$X \rightarrow Y, \quad (x, y) \mapsto (x^p - x, y)$$

is H -equivariant and of degree p , which implies that it is the quotient map $X \rightarrow X/H$. The action of C on Y is given by

$$\rho(z) = \zeta^{-m} \cdot z, \quad \rho(y) = \zeta^{-1} \cdot y.$$

The following lemma is crucial in the proof of Proposition 7.1.

Lemma 7.2. *Keep the above notation. Then, as $k[H]$ -modules:*

$$\begin{aligned}H^0(Y, \Omega_Y) &\cong \bigoplus_{i=0}^{m \cdot (p-1)-1} (\psi^i)^{\oplus \alpha_i}, \\ H^0(Y, \Omega_Y(D)) &\cong \bigoplus_{i=0}^{m \cdot (p-1)-1} (\psi^i)^{\oplus \beta_i}, \\ H^0(Y, \Omega_Y(R^l)) &\cong \bigoplus_{i=0}^{m \cdot (p-1)-1} (\psi^i)^{\oplus \gamma_i}.\end{aligned}$$

Proof. Similarly as above, one checks that the quotient map $Y \rightarrow Y/C \cong \mathbb{P}^1$ is given by $(y, z) \mapsto z^{p-1}$. The set of ramification points of $Y \rightarrow Y/C$ is given by $\{Q_\infty, Q_0, Q_1, \dots, Q_N\} \subset (Y/C)(k)$, where $N := \frac{p^{n-1}-1}{p-1}$, $Q_0 = 0$, $Q_\infty = \infty$ and $Q_1, \dots, Q_N$ are the elements of the set

$$\{\bar{v}^{p-1} : \bar{v} \in \bar{V} \setminus \{0\}\} \subset (Y/C)(k).$$

One easily checks that:

$$C_{Q_i} = \begin{cases} C, & \text{for } i = 0, \infty, \\ C', & \text{for } i = 1, \dots, N, \end{cases}$$

where $C' := \langle \rho^{p-1} \rangle \cong \mathbb{Z}/m$. The branch points of $\pi : Y \rightarrow Y/C$ are the points of Y given as follows:

- points P_0 and P_∞ above Q_0 and Q_∞ respectively,
- points $P_i^{(1)}, \dots, P_i^{(p-1)}$ above Q_i for $i = 1, \dots, N$.

Moreover, $B_{X/Y} = \{P_\infty\}$ and $u_{X/Y, P_\infty} = m$. Note that y is the uniformizer of points $P_i^{(j)}$. Thus, since $\rho(y) = \zeta^{-1} \cdot y$, we see that for $0 \leq i \leq N$, $1 \leq t \leq e_{P_i^{(j)}} - 1$:

$$\begin{aligned} \theta_{Y/\mathbb{P}^1, Q_i} &= \begin{cases} \psi^{-1}, & \text{if } i = 0, \\ (\psi')^{-1}, & \text{otherwise,} \end{cases} \\ N_{Q_i, t}(\psi^l) &= \begin{cases} \llbracket -t \equiv l \pmod{m \cdot (p-1)} \rrbracket, & \text{if } i = 0, \\ \llbracket -t \equiv l \pmod{m} \rrbracket, & \text{otherwise,} \end{cases} \end{aligned}$$

where $\psi' := \psi|_{C'}$. Moreover, since the uniformizer at P_∞ is given by x^{c_1}/y^{c_2} , we obtain

$$\theta_{Y/\mathbb{P}^1, Q_\infty} = \psi^{c_2 - m \cdot c_1}, \quad N_{Q_\infty, t}(\psi^l) = \llbracket (c_2 - m \cdot c_1) \cdot t \equiv l \pmod{m \cdot (p-1)} \rrbracket.$$

Therefore, by Proposition 2.1, the multiplicity of the character ψ^l in the $k[H]$ -module $H^0(X, \Omega_X)$ equals:

$$\begin{aligned} (g_{\mathbb{P}^1} - 1) + \sum_{j=1}^N \sum_{t=1}^{m-1} \left\langle \frac{-t}{m} \right\rangle \cdot N_{Q_j, t}(\psi^l) + \sum_{t=1}^{(p-1) \cdot m-1} \left\langle \frac{-t}{(p-1) \cdot m} \right\rangle \cdot N_{Q_0, t}(\psi^l) \\ + \sum_{t=1}^{(p-1) \cdot m-1} \left\langle \frac{-t}{(p-1) \cdot m} \right\rangle \cdot N_{Q_\infty, t}(\psi^l) + \llbracket l = 0 \rrbracket. \end{aligned}$$

This expression is easily seen to be equal to α_l . The formulas for $H^0(Y, \Omega_Y(D))$ and $H^0(Y, \Omega_Y(R'))$ follow analogously from Proposition 2.1. $\square$

Proof of Proposition 7.1. By Theorem 6.1 we have $H_{dR}^1(X) \cong J_p(V_1) \oplus J_{p-1}(V_2)$, where (using Lemma 7.2):

$$\begin{aligned} V_1 &\cong H^0(Y, \Omega_Y) \oplus H^1(Y, \mathcal{O}_Y) \cong \bigoplus_{l=0}^{m \cdot (p-1)} (\psi^l)^{\oplus(\alpha_l + \alpha_{-l})} \\ V_2 &\cong H^0(Y, \Omega_Y(R')) \oplus H^1(Y, \mathcal{O}_Y(-D))^{x^{-1}} \oplus H^1(Y, \mathcal{O}_Y)^{x^{-1}} \oplus H^0(Y, \Omega_Y) \\ &\cong \bigoplus_{l=0}^{m \cdot (p-1)} (\psi^l)^{\oplus(\gamma_l + \beta_{-l-m} - \alpha_{-l+m} - \alpha_l)}. \end{aligned}$$

The result follows. $\square$

Data availability

No data was used for the research described in the article.

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